

MicroCODE Software Engineering Services

Project Coding Standards

MCX-S02 (Internal JS Style Guide) v008.docx

Development Environment: Microsoft Visual Studio Code

OS: HTML5 Browser Environment

Platform: Open Source ES6, HTML5, CSS3

Language: JavaScript (ECMAScript 6 – ES6)

Tools: JSLint, JSHint

MicroCODE JavaScript Style Guide

This was adopted from the **MIT xPRO JavaScript Certificate Style Guide**, differences are noted. The MIT guide was in turn adapted from the [Airbnb style guide](#) and [the Turing School style guide](#).

Everything we believe is based on this statement:

Code like a Machine: Consistently and Explicitly, Simply, and for Readability (Hail CAESAR!).

All our rules are marked with one or more of these to relate back to this statement:

(Code Consistently) (Code Explicitly) (Code Simply) (Code for Readability)

Table of Contents

Standard Terminology – MicroCODE additions

Standard JSDocs – MicroCODE additions

Font Selection and Readability – MicroCODE addition

Indents, Braces, Blocks, and Aligned-Code – MicroCODE addition

Whitespace

Types

Objects

Arrays

Strings

Functions

Arrow Functions – MicroCODE addition

Properties

Variables

Comparison Operators & Equality

Blocks

Commas

Semicolons

Naming Conventions

Common Patterns – MicroCODE addition

Appendix A: JS Class Structure – MicroCODE addition

Appendix B: VS Code Settings – MicroCODE addition



Standard Terminology (MicroCODE)

Any Style Guide should start with **coding terminology** because there is a general lack of standardized terms.

() – these are **PARENS**, short for parentheses, they are not ‘round brackets’ or ‘round braces’, parens are, by our definition, a pair of round delimiters.

[] – these are **BRACKETS**. They are not braces, they are not ‘square brackets’, brackets are, by our definition, a pair of square delimiters.

{ } – these are **BRACES**. They are not brackets, they are not ‘curly brackets’ nor ‘curly braces’, braces are by our definition a pair of a ‘curly’ delimiters. Saying ‘curly braces’ is redundant, saying ‘curly brackets’ is just wrong.

< > – these are **ANGLES**. They are not brackets, they are not ‘angled brackets’ nor ‘angled braces’, angles are, by our definition, a pair of angular delimiters.

In typography all of these are referred to generically as ‘brackets’. A nice article explaining this: <https://type.today/en/journal/brackets>. But we are not talking about typesetting, we are talking about coding, and precise language eliminates confusion and mistakes; it produces consistent results and saves time & money.

And so, for all our code and documentation... ([Code Consistently](#))

Code Delimiters

```
( ) = PARENS:   parameter grouping, operand precedence, quantities
[ ] = BRACKETS: indexing, array formation
{ } = BRACES:   code blocks, initialization values (compiler)
< > = ANGLES:  substitution identifier, option grouping
```

Boolean Expressions

```
& = AND: inclusive
| = OR:  optional
! = NOT: exclusive

() = AND Group, everything within is AND'ed
<> = OR Group, everything within is OR'ed
```

Boolean Equations

```
RPO ABC <XYZ ABO YUT> !ZZY Z71
RPO ABC <XYZ (ABO BBC !CDC) YUT> !ZZY Z71
```

Implies, in more detail...

```
&RPO &ABC <XYZ ABO YUT> !ZZY &Z71
&RPO &ABC <XYZ (ABO BBC !CDC) YUT> !ZZY &Z71
```

Implies, in complete notation...

```
&RPO &ABC &<XYZ|ABO|YUT> &!ZZY &Z71
&RPO &ABC &<XYZ|(&ABO &BBC &!CDC)|YUT> &!ZZY &Z71
```



Standard Terminology (continued)

Interlocks, Signals, and Memories vary in duration, persistence, and interactions. This should be understood immediately by the 'Parameter Persistence Type'. ([Code Consistently](#))

ONE-SHOT – an application variable—a BOOLEAN—that goes from FALSE to TRUE for a single program scan or routine execution.

RISING-EDGE – a signal condition—a BOOLEAN—that indicates the variable has gone from FALSE to TRUE during this program scan or routine execution.

FALLING-EDGE – a signal condition—a BOOLEAN—that indicates the variable has gone from TRUE to FALSE during this program scan or routine execution.

PULSE – an application variable—a BOOLEAN—that goes from FALSE to TRUE for specified (or configured) duration, normally specified in milliseconds.

FLIP – a variable operation—performed on a BOOLEAN—that reverses it from TRUE to FALSE or FALSE to TRUE based on its initial state.

LATCH – a variable operation—performed on a BOOLEAN—that sets it to TRUE regardless of its initial state.

UNLATCH – a variable operation—performed on a BOOLEAN—that sets it to FALSE regardless of its initial state.

INTERLOCK – an application variable—any PRIMITIVE or OBJECT—that is presented to external equipment or a remote application until it is Acknowledged (ACK/NAK). This implies retries and failure annunciation if not acknowledged.

MEMORY – an application variable—any PRIMITIVE or OBJECT—that changes state based on program scan or routine execution. These can be *temporary* or *persistent*.

CONTEXT – the sum total of all the application's **Persistent Memory** that controls its behavior. Persistent Memory should be stored permanently from one session to another for a consistent User Experience (UX). The quality of this persistence is what gives a user confidence in a system or application, providing the feeling that 'it remembers what I was doing, and I did not have to configure anything for that to happen'.

CONFIGURATION – a special subset of the application's **Persistent Memory** that represents User Preferences or User Settings, usually configured by them thru the User Interface (UI) thru the 'Gear' or 'Cog' Icon. These settings must be protected thru all Application Upgrades as they represent 'User Work Products'. These settings must be written to permanent data storage, i.e.: a configuration file or database.



Standard JSDocs (MicroCODE)

Code should be documented, period. There's a crazy idea out there that 'documentation clutters the code', or 'documentation takes too much time', or, even crazier... 'code should contain no documentation' (actuality espoused by an IT person in General Motors). For anyone who has built an application of any significance, one that lasts for years, or decades in the field, knows that **you** the author will most likely need that documentation years from now when an issue arises. ([Code for Readability](#))

So, our rules are simple:

- 1) Use **JSDocs** headers for all major functions, classes, data types, or infrastructure.
- 2) Add inline comments where the function—or side-effects—of the code are not obvious.

JSDocs Header should contain, in this order...

- 1) **@function** (or **@method**) – Indicates that the comment is documenting a function or method. This can be especially important in codebases where the distinction between functions and other types of declarations is emphasized in the documentation.
- 2) **@memberof** – Specifies the parent object or module that the function or variable is part of. This is essential for organizing the documentation in a hierarchical structure and making it clear how different parts of the application relate to each other.
- 3) **@description** – Provides a detailed description of the function or variable. This can also be the implicit description block at the beginning of the comment.
- 4) **@param** (or **@arg**, **@argument**) – Documents the parameters that a function takes.
- 5) **@returns** (or **@return**) – Describes the value returned by the function.
- 6) **@example** – Shows usage examples for the function or variable.
- 7) **@see** - References related documentation.
- 8) Accessibility tags like **@public**, **@private**, **@protected** to indicate the visibility or accessibility level.
- 9) **@deprecated** – Notes that the documented item is deprecated.
- 10) **@since** – indicates the version when an item was added.
- 11) Additional tags like **@todo**, **@throws** (or **@exception**), **@type**, and others as needed.

Including **@function** and **@memberof** at the top emphasizes the structural aspects of the documentation, making it clearer from the outset what is being documented and where it fits within the project's architecture.



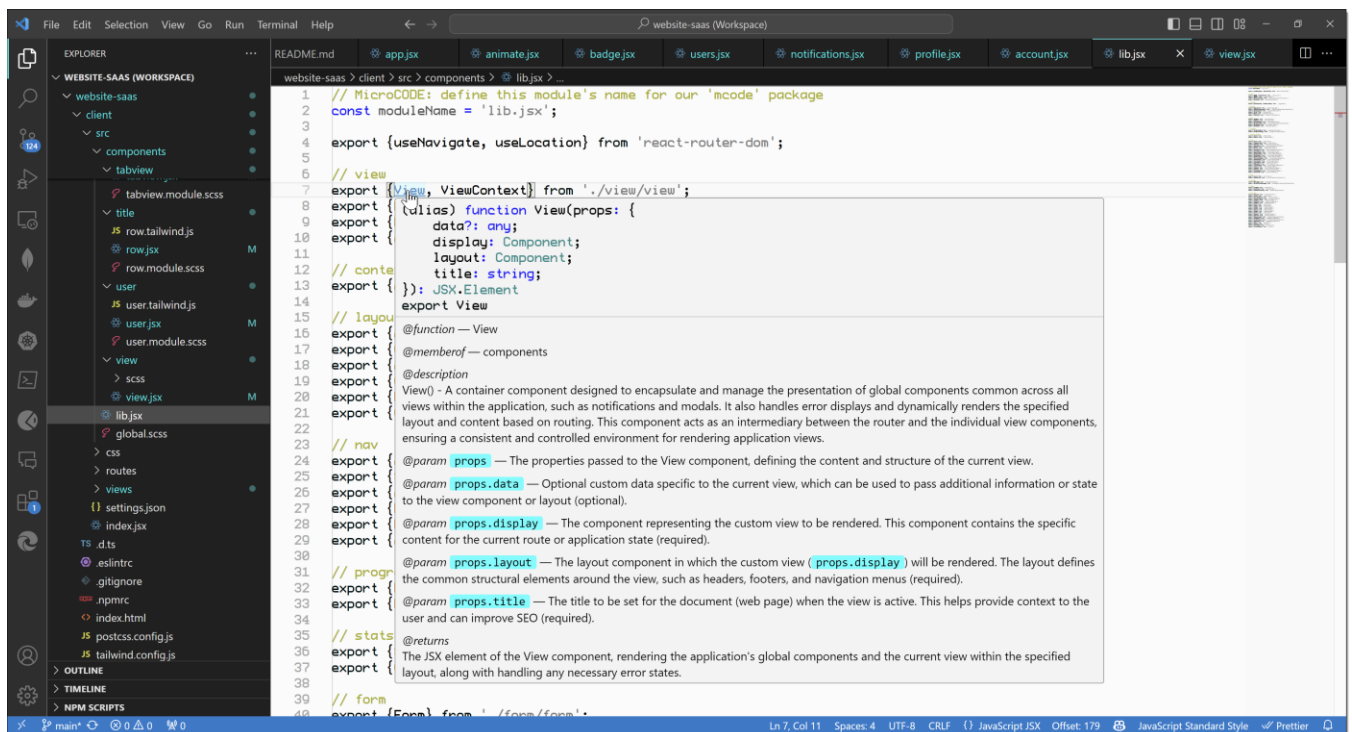
Example 1 Basic JSDocs Header for a Component JSX Function

```

/**
 * @function View
 * @memberof components
 * @description
 * View() - A container component designed to encapsulate and manage the presentation of global components
 * common across all views within the application, such as notifications and modals. It also handles error
 * displays and dynamically renders the specified layout and content based on routing. This component acts
 * as an intermediary between the router and the individual view components, ensuring a consistent and
 * controlled environment for rendering application views.
 *
 * @param {Object} props - The properties passed to the View component, defining the content and structure of the
 * @param {Object} [props.data] - Optional custom data specific to the current view, which can be used to pass ad
 * @param {Component} props.display - The component representing the custom view to be rendered. This component c
 * @param {Component} props.layout - The layout component in which the custom view ('props.display') will be rend
 * @param {string} props.title - The title to be set for the document (web page) when the view is active. This he
 *
 * @returns {JSX.Element} The JSX element of the View component, rendering the application's global components
 * and the current view within the specified layout, along with handling any necessary error states.
 */
1 reference
export function View(props)
{
  const navigate = useNavigate();

```

Example 2 VS Code 'IntelliSense' display from JSDocs



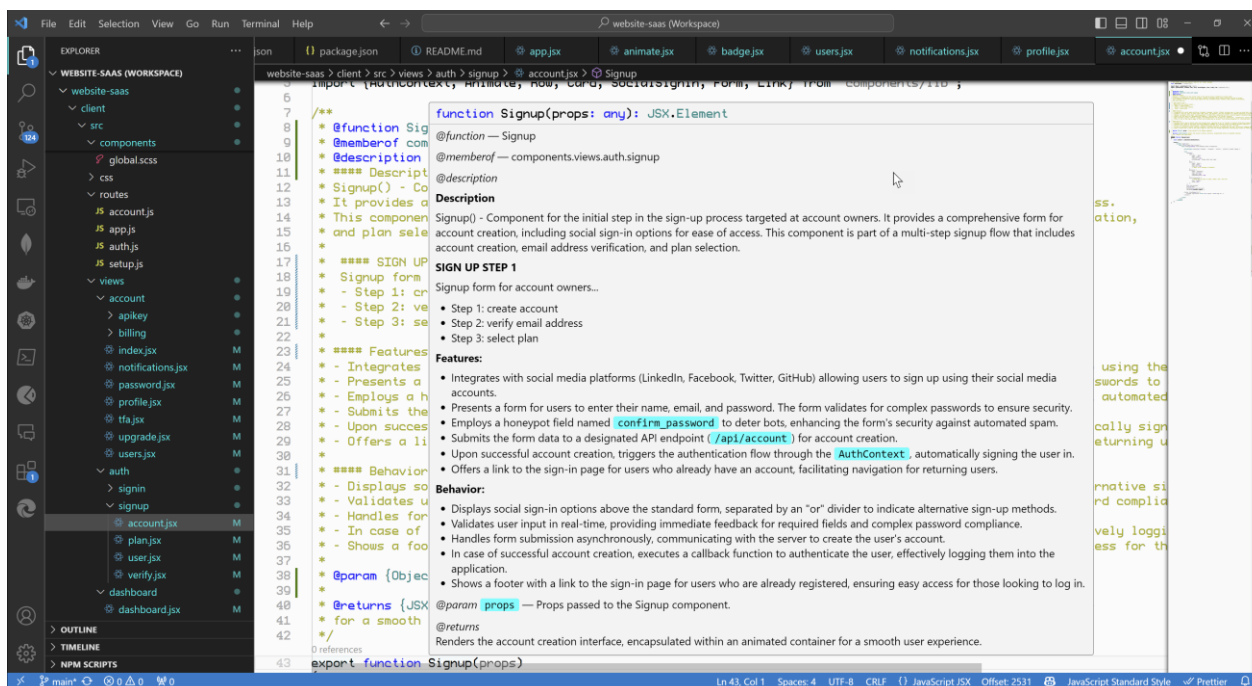
Example 3 Detailed JSDocs Header for a High-Level Function

```

/**
 * @function Signup
 * @memberof components.views.auth.signup
 * @description
 * ##### Description
 * Signup() - Component for the initial step in the sign-up process targeted at account owners.
 * It provides a comprehensive form for account creation, including social sign-in options for ease of access.
 * This component is part of a multi-step signup flow that includes account creation, email address verification,
 * and plan selection.
 *
 * ##### SIGN UP STEP 1
 * Signup form for account owners...
 * - Step 1: create account
 * - Step 2: verify email address
 * - Step 3: select plan
 *
 * ##### Features:
 * - Integrates with social media platforms (LinkedIn, Facebook, Twitter, GitHub) allowing users to sign up using th
 * - Presents a form for users to enter their name, email, and password. The form validates for complex passwords to
 * - Employs a honeypot field named 'confirm_password' to deter bots, enhancing the form's security against automate
 * - Submits the form data to a designated API endpoint ('/api/account') for account creation.
 * - Upon successful account creation, triggers the authentication flow through the 'AuthContext', automatically sig
 * - Offers a link to the sign-in page for users who already have an account, facilitating navigation for returning
 *
 * ##### Behavior:
 * - Displays social sign-in options above the standard form, separated by an "or" divider to indicate alternative s
 * - Validates user input in real-time, providing immediate feedback for required fields and complex password compli
 * - Handles form submission asynchronously, communicating with the server to create the user's account.
 * - In case of successful account creation, executes a callback function to authenticate the user, effectively logg
 * - Shows a footer with a link to the sign-in page for users who are already registered, ensuring easy access for t
 *
 * @param {Object} props - Props passed to the Signup component.
 *
 * @returns {JSX.Element} Renders the account creation interface, encapsulated within an animated container
 * for a smooth user experience.
 */
0 references

```

Example 4 VS Code 'IntelliSense' display from JSDocs

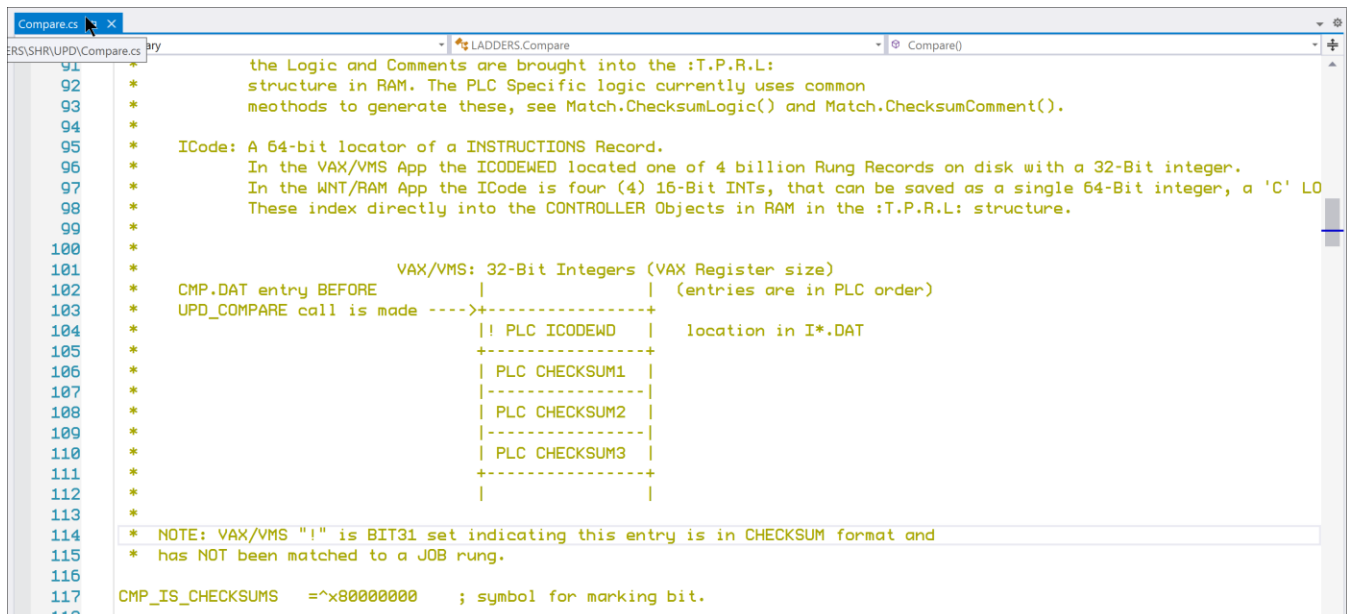


Font Selection and Readability (MicroCODE)

We believe you should always code in a mono-spaced font. (Code Consistently)

The only explanation required is a couple simple examples:

Monospace – enables the creation of text tables in documentation.

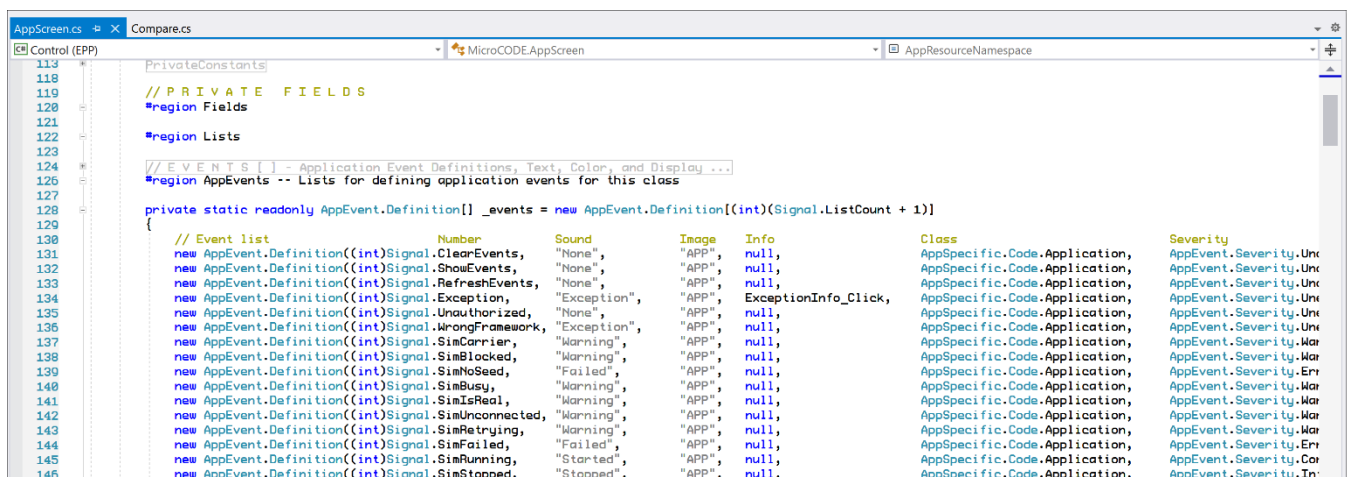


```

Compare.cs
LADDERS.Compare
Compare()

91 * the Logic and Comments are brought into the :T.P.R.L:
92 * structure in RAM. The PLC Specific logic currently uses common
93 * methods to generate these, see Match.ChecksumLogic() and Match.ChecksumComment().
94 *
95 * ICode: A 64-bit locator of a INSTRUCTIONS Record.
96 * In the VAX/VMS App the ICODEWED located one of 4 billion Rung Records on disk with a 32-Bit integer.
97 * In the WNT/NT App the ICode is four (4) 16-Bit INTs, that can be saved as a single 64-Bit integer, a 'C' LO
98 * These index directly into the CONTROLLER Objects in RAM in the :T.P.R.L: structure.
99 *
100 *
101 * VAX/VMS: 32-Bit Integers (VAX Register size)
102 * CMP.DAT entry BEFORE |-----| (entries are in PLC order)
103 * UPD_COMPARE call is made ---->+-----+
104 * |! PLC ICODEWD | location in I*.DAT
105 * +-----+
106 * | PLC CHECKSUM1 |
107 * +-----+
108 * | PLC CHECKSUM2 |
109 * +-----+
110 * | PLC CHECKSUM3 |
111 * +-----+
112 *
113 *
114 * NOTE: VAX/VMS "I" is BIT31 set indicating this entry is in CHECKSUM format and
115 * has NOT been matched to a JOB rung.
116
117 CMP_IS_CHECKSUMS = ^x80000000 ; symbol for marking bit.
118
  
```

Monospace – makes coded tables instantly readable and exceptions pop-out.



```

AppScreen.cs
Compare.cs
AppResourceNamespace

113 * Private Constants
114 *
115 * // PRIVATE FIELDS
116 * #region Fields
117 *
118 * #region Lists
119 *
120 * // EVENTS [ ] - Application Event Definitions, Text, Color, and Display ...
121 * #region AppEvents -- Lists for defining application events for this class
122 *
123 * private static readonly AppEvent.Definition[] _events = new AppEvent.Definition[(int)(Signal.ListCount + 1)]
124 * {
125 *     // Event list
126 *     new AppEvent.Definition((int)Signal.ClearEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
127 *     new AppEvent.Definition((int)Signal.ShowEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
128 *     new AppEvent.Definition((int)Signal.RefreshEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
129 *     new AppEvent.Definition((int)Signal.Exception, "Exception", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
130 *     new AppEvent.Definition((int)Signal.Unauthorized, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
131 *     new AppEvent.Definition((int)Signal.WrongFramework, "Exception", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
132 *     new AppEvent.Definition((int)Signal.SimCarrier, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unk),
133 *     new AppEvent.Definition((int)Signal.SimBlocked, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.War),
134 *     new AppEvent.Definition((int)Signal.SimNoSeed, "Failed", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Err),
135 *     new AppEvent.Definition((int)Signal.SimBusy, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.War),
136 *     new AppEvent.Definition((int)Signal.SimIsReal, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.War),
137 *     new AppEvent.Definition((int)Signal.SimUnconnected, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.War),
138 *     new AppEvent.Definition((int)Signal.SimRetrying, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.War),
139 *     new AppEvent.Definition((int)Signal.SimFailed, "Failed", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Err),
140 *     new AppEvent.Definition((int)Signal.SimRunning, "Started", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Cor),
141 *     new AppEvent.Definition((int)Signal.SimStopped, "Stopped", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.In)
142 * }
143
  
```



Proportional – creates a mess. A must for User documentation, bad for code.

```

191 * the Logic and Comments are brought into the :T.P.R.L:
192 * structure in RAM. The PLC Specific logic currently uses common
193 * methods to generate these, see Match.ChecksumLogic() and Match.ChecksumComment().
194 *
195 * ICode: A 64-bit locator of a INSTRUCTIONS Record.
196 * In the VAX/VMS App the ICODEWED located one of 4 billion Rung Records on disk with a 32-Bit integer.
197 * In the WNT/NT App the ICode is four (4) 16-Bit INTs, that can be saved as a single 64-Bit integer, a 'C' LONG.
198 * These index directly into the CONTROLLER Objects in RAM in the :T.P.R.L: structure.
199 *
200 *
201 * VAX/VMS: 32-Bit Integers (VAX Register size)
202 * CMP.DAT entry BEFORE | | (entries are in PLC order)
203 * UPD_COMPARE call is made --> +-----+
204 * |! PLC ICODEWD | location in I*.DAT
205 * +-----+
206 * | PLC CHECKSUM1 |
207 * +-----+
208 * | PLC CHECKSUM2 |
209 * +-----+
210 * | PLC CHECKSUM3 |
211 * +-----+
212 * | |
213 *
214 * NOTE: VAX/VMS "I" is BIT31 set indicating this entry is in CHECKSUM format and
215 * has NOT been matched to a JOB rung.
216 *
217 CMP_IS_CHECKSUMS =^x80000000 : symbol for marking bit.
218 *

```

Proportional – creates a mess, coded tables are much harder to read.

```

120 #region Fields
121
122 #region Lists
123
124 // EVENTS[] - Application Event Definitions, Text, Color, and Display...
125 #region AppEvents - Lists for defining application events for this class
126
127 private static readonly AppEvent.Definition[] _events = new AppEvent.Definition[(int)(Signal.ListCount + 1)]
128 {
129     // Event list
130     Number Sound Image Info Class Severity Audience Repeatable Persistent Targets
131     new AppEvent.Definition((int)Signal.ClearEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Undefined, AppEvent.Audience.Undefined, f,
132     new AppEvent.Definition((int)Signal.ShowEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Undefined, AppEvent.Audience.Operator, f,
133     new AppEvent.Definition((int)Signal.RefreshEvents, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Undefined, AppEvent.Audience.Operator, f,
134     new AppEvent.Definition((int)Signal.Exception, "Exception", "APP", ExceptionInfo_Click, AppSpecific.Code.Application, AppEvent.Severity.Unexpected, AppEvent.Audience.Develop,
135     new AppEvent.Definition((int)Signal.Unauthorized, "None", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unexpected, AppEvent.Audience.Develop,
136     new AppEvent.Definition((int)Signal.WrongFramework, "Exception", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Unexpected, AppEvent.Audience.Develop,
137     new AppEvent.Definition((int)Signal.SimCarrier, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Warning, AppEvent.Audience.Operator, tr,
138     new AppEvent.Definition((int)Signal.SimBlocked, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Warning, AppEvent.Audience.Operator, ti,
139     new AppEvent.Definition((int)Signal.SimNoSeed, "Failed", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Error, AppEvent.Audience.Operator, true,
140     new AppEvent.Definition((int)Signal.SimBusy, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Warning, AppEvent.Audience.Operator, tr,
141     new AppEvent.Definition((int)Signal.SimIsReal, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Warning, AppEvent.Audience.Operator, tr,
142     new AppEvent.Definition((int)Signal.SimUnconnected, "Warning", "APP", null, AppSpecific.Code.Application, AppEvent.Severity.Warning, AppEvent.Audience.Operator,

```



[illegible]

```

113 PrivateConstants
118
119 // PRIVATE FIELDS
120 *region Fields
121
122 *region Lists
123
124 // EVENTS[] - Application Event Definitions, Text, Color, and Display ...
126 *region AppEvents -- Lists for defining application events for this class
127
128 private static readonly AppEvent.Definition[] _events = new AppEvent.Definition[(int)(Signal.ListCount + 1)]
129 {
130     // Event list
131     new AppEvent.Definition((int)Signal.ClearEvents, "None", "APP", null, AppSpec
132     new AppEvent.Definition((int)Signal.ShowEvents, "None", "APP", null, AppSpec
133     new AppEvent.Definition((int)Signal.RefreshEvents, "None", "APP", null, AppSpec
134     new AppEvent.Definition((int)Signal.Exception, "Exception", "APP", ExceptionInfo_Click, AppSpec
135     new AppEvent.Definition((int)Signal.Unauthorized, "None", "APP", null, AppSpec
136     new AppEvent.Definition((int)Signal.WrongFramework, "Exception", "APP", null, AppSpec
137     new AppEvent.Definition((int)Signal.SimCarrier, "Warning", "APP", null, AppSpec
138     new AppEvent.Definition((int)Signal.SimBlocked, "Warning", "APP", null, AppSpec

```

```

0 1 2 3 4 5 6 7 8 9 0
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
@ # $ % ^ & * ( ) [ ] { } < > _ + - = \ / |
; : ' " , . ? ! ` ~

```

Note: [11:16:55 AM 24-Oct-2023] Update. Google has released a great coding font named: “Fira Code”

Link: <https://fonts.google.com/specimen/Fira+Code?query=fira+code>

This font was designed for coding and specifically for drawing pictures—as referenced above—in your code or monospace output.

LADDERS Compare Report example:

The bar graph, arrow lines, all are just ASCII rendered with **Fira Code** font...

MicroCODE Incorporated
LADDERS®
Compare Report
App Version v1.0.0 Beta 4

Allen-Bradley Logix 5000
EP_GTG01Z1
EP_GTG01Z1_CDE_REV05.L5K (OLD)
EP_GTG01Z1_HTI_20230927_0850AM.L5K (NEW)

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<https://www.mcode.com>
Report No.: **LDRS230930.1155.06**



NEW PLC: Lines (RUNGs)

NEW RUNGs:	COMPARISON:	OLD RUNGs:	CODE EDIT LEVEL:
			0 25 50 75 100
	DELETED →	2	
	DELETED →	3	
	DELETED →	4	
	DELETED →	5	
Routine: Prescan (I) (QDI Event Record)			
	DELETED →	0	
TASK: MainTask (I+C)			
PROGRAM: MCP (I+C) (A000_Main)			
ROUTINE: A000_Main (I) (Main Control Rou)			
1	← INSERTED		
2	← INSERTED		

LINEs Summary

LINEs of Code (L)	
TOTAL TASKs in NEW PLC	5
TASKs INSERTED in NEW	0
TOTAL PROGRAMs in NEW PLC	13
PROGRAMs INSERTED in NEW	5
TOTAL ROUTINEs in NEW PLC	511
ROUTINEs INSERTED in NEW	47
TOTAL RUNGs in NEW PLC	7056
RUNGs INSERTED in NEW	369
RUNGs that matched EXACTLY	6515
RUNGs (I) that were EDITED	172
RUNG COMMENTS (C) EDITED	53
RUNGs DELETED from OLD	615

While we still prefer **ProFont** for coding, this is essential for the LADDERS output, and we have embedded the font directly into the report output.

Indents, Braces, Blocks, and Aligned-Code (MicroCODE)

Common JavaScript Style Guides like to use Kernighan & Richie (K&R) 'Egyptian Brackets'.

Right off the bat, they are really not talking about “brackets” they are talking about “braces”.

Our opinion is that this style ‘saves’ one line break per block of code at the expense of readability.

Why? Because the key word expressions like ‘if’ and ‘else’ overlap and offset the code of the clause.

This style saved screen space when people coded on 24-Line CRTs (like the VT52s, VT100s, VT220s, etc.) and saved paper when people printed code. We have neither of these limitations in the 21st Century; white space is your friend, delimiter alignment is your friend, language keyword alignment is your friend.

```
if (condition) {  
    // true code  
} else {  
    // false code  
}  
  
try {  
    // protected code  
} catch {  
    // correction code  
} finally {  
    // exit code  
}
```

K&R Style

The above is **K&R**, below is **BSD** which is our preferred JavaScript Style (and C/C++/C# Style as well).

The braces are aligned.

The key words are aligned.

All code blocks are held within aligned braces.

All conditional code has natural white space around it via the braces.

People will argue that K&R is easy to read, but they are relaying on colorized text editors that are hiding the readability issue in the bare text. **There is no world in which K&R is easier to read and maintain than BSD.**

Note: Aligned-Code is also known as **Allman Style** (after Eric Allman) or **BSD Style** – Berkeley Software Distribution.

```
if (condition)  
{  
    // true code  
}  
else  
{  
    // false code  
}  
  
try  
{  
    // protected code  
}  
catch  
{  
    // correction code  
}  
finally  
{  
    // exit code  
}
```

BSD Style



Notice in the second example the conditional code is naturally separated by white space created by the balanced braces, this makes the code far more readable than the 'Egyptian Brackets'. And there are no excuses based on typing speed, as these preferences can all be enforced automatically by VS CODE Settings.

White space is your friend and aids in readability; half the effort of maintaining code is readability and consistency.

As an extension, when building conditional execution, braces should always be used—and all code blocks should be placed on the lines following the conditional expression.

```
if (condition) ... // true code - DO NOT DO THIS, see exception on next page
if (condition)
  ... // true code - DEFINITELY DO NOT DO THIS
if (condition)
{
  ... // true code - ALWAYS DO THIS, conditional code in a BLOCK of BRACES, 'ALIGNED'
  ... // here is example a line added in the future, no reformatting required
}
```

This first reason for this is consistency and long-term maintenance. If the first two examples ever need editing—where an additional line of code needs to be added to the execution clause—the first thing that must be done is the addition of braces and the reformatting of the lines... a burden placed on the future coder by the original author. If line(s) of code are added in a hurry—and no braces are added—you just cost the next coder (or your future self) a good :10 minutes figuring out why their condition code is not working. It just not worth it.

The second reason is readability, it's just far easier to read all conditional code the same way, everywhere in a project, if it is all formatted exactly the same way.

It is often said: "Cleanliness is Next to Godliness", in coding we believe "Consistency is Next to Godliness".

The universe works on a set of consistent rules for a reason. If you code consistently—always following the same rules, always following the same patterns, where all code looks like it came from the same author, where all Classes follow the same format—you and your Team will reap the benefits now and forever. If your code is utterly consistent your Team members (or an A.I. Assistant like **ChatGPT** or GitHub **Copilot**) can code with predictions that will always be true... **that** will speed up development more than any style can by saving typing white space or braces.

See **Appendix A: MicroCODE JS Class Structure** as an example of using a Template to make following these rules simple.

Our Mantra: "**Code like a Machine: Consistently and Explicitly, Simply and for Readability** (Hail CEASAR)."SM

The combination of BSD Style, with the MCODE Rules, like always starting new modules from approved templates—no matter how simple the new Class or Module may be—we call...

MCODE Style

We've marked all AirBnB Rules that are eliminated as a side-effect of MCODE Style with...

MCODE Style eliminates



There are always exceptions to rules and here's the one for our use of BSD Style.

It is quite common in JavaScript to do fast checks for an 'early exit', for readability and performance.

Those tend to be very simple one-liners like this...

```
static #applySensorState(element, state)
{
    if (!element) return; // ALWAYS DO THIS for simple 1-line statements
    ...                  // where there is no chance of this code expanding

static #applyLed(element, state)
{
    if (!element)
    {
        return; // - DO NOT DO THIS
    }
}
```

...use best judgment, keep the code clean, simple, and as compact as possible without losing readability or comprehension.



Whitespace

- Use soft tabs set to ~~2 spaces~~ 4 spaces. (Code for Readability).
- **Note:** Use 8 spaces for Assembly Language files from 1980s and 1990s or you will destroy diagrams, tabular data, etc. as these were created in CRT / VT100 environments.

```
// bad
function foo() {
var name;
}

// bad
function bar() {
var name;
}

// good
function baz()
{
var name;
}
```

MCODE Style eliminates

- ~~Place 1 space before the leading curly brace.~~
- Start all code blocks on new lines with aligned braces. (Code Consistently, Code for Readability).

```
// bad
function test(){
console.log('test');
}

// good
function test()
{
console.log('test');
}
```

MCODE Style eliminates

- ~~Place 1 space before the opening parenthesis in control statements (if, else if, while etc.).~~
- Start all code blocks on new lines with aligned braces. (Code Consistently, Code for Readability).

```
// bad
if(isSithLord) {
fight();
}

// good
if (isSithLord)
{
fight();
}
```

MCODE Style eliminates



- Place no space before the argument list in function calls and declarations. ([Code Consistently](#)).

```
// bad
function fight () {
  console.log ('Swoooosh!');
}

// good
function fight()
{
  console.log('Swoooosh!');
}
```

MCODE Style eliminates

- Set off operators with spaces.

```
// bad
var x=y+5;

// good
var x = y + 5;
```

- Leave a blank line after blocks and before the next statement. ([Code Consistently](#), [Code for Readability](#)).
- **Note:** This rule makes no sense when 'Egyptian Brackets' are used.
- **Never place more than one blank line between code lines of any type.**

```
// bad
if (foo) {
  return bar;
}
return baz;

// good
if (foo)
{
  return bar;
}

return baz;

// bad
var obj = {
  foo: function () {},
  bar: function () {}
};
return obj;

// good
var obj =
{
  foo: function () {},
  bar: function () {}
};

return obj;
```

MCODE Style eliminates



Types

- Use native data types for any variables that are not application specific or persistent. ([Code Simply](#)).

```
// bad
var x = {};
x.counter = 0;
for (x.counter = 0; x.counter < 99; x.counter++)
{
    // work
}

// good
for (let counter = 0; counter < 99; counter++)
{
    // work
}
```

- Use the simplest data structures possible. ([Code Simply](#)).



Objects

- Use the literal syntax for generic object creation. ([Code Simply](#)).

```
// bad
var item = new Object();

// good
var item = {};
```

- Don't use [reserved words](#) as keys. It won't work in IE8. [More info](#). ([Code Explicitly](#)).

```
// bad
var superman = {
  default: { clark: 'kent' },
  private: true
};

// good
var superman =
{
  alterEgo: { clark: 'kent' },
  hidden: true
};
```

- Use readable synonyms in place of reserved words. ([Code Explicitly](#)).

```
// bad
var superman = {
  class: 'alien'
};

// bad
var superman = {
  klass: 'alien'
};

// good
var superman =
{
  species : 'kryptonian'
};
```



Arrays

- Use the literal syntax for array creation. ([Code Simply](#)).

```
// bad
var items = new Array();

// good
var items = [];
```

- Use `Array.push` instead of direct assignment to add items to an array. ([Code Simply](#)).

```
var someStack = [];

// bad
someStack[someStack.length] = 'bohemianrhapsody';

// good
someStack.push('bohemianrhapsody');
```

- `.push` (add to end), `.pop` (remove from end) work on the **end** of the array
- `.unshift` (add to start), `.shift` (remove from start) work on the **start** of the array
- **Note:** `.shift` SHIFTS LEFT, `.unshift` SHIFTS RIGHT



Strings

- Use single quotes `'` for most strings, even though double-quotes `"` are allowed. ([Code Consistently](#)).
- Use double quotes `"` for string containing single quotes, or escape `'` in strings as `\'`.
- Use back-ticks (grave accents) ``` for template literals or template strings. ([JavaScript language](#)).

```
// preferred - no quotes
var name = 'Bob Barker';

// ok - with apostrophes
var name = "Bob's Books";

// preferred
var name = 'Bob\'s Books';

// bad
var fullName = "Bob " + this.lastName;

// preferred, and JS required
var fullName = `Bob ${this.lastName}`;
```

- Use single quotes `'` for string holding double-quotes (HTML, JSON). ([Code Consistently](#)).
- Why? Because JavaScript functions are often building HTML or CSS code that has embedded double-quoted values which must be double-quoted.

```
ui.result =
  <div class="card text-white bg-success mb-3" style="max-width: 40rem;">
    <div class="card-header" id="resultTitle">Result Display</div>
    <div class="card-body">
      <h5 class="card-title">Transaction Completed</h5>
      <label for="outputResult" class="col-sm-40 col-form-label">Result</label>
      <div class="col-sm-40">
        <input type="text" id="outputResult" value="<result>">
      </div>
    </div>
  </div>;
```



Table 1 JavaScript String (JSS)

JavaScript Strings and Delimiters	
Use Case	Preferred Quotes
JavaScript Strings (JSS)	' ' (single quotes – simplest JavaScript, outer HTML)
JSON definitions within JSS	" " (double quotes – JSON file format)
HTML strings within JSS	" " (double quotes – HTML format)
SQL Queries within JSS	' ' (single quotes – contain many " " parameters)
JS Dynamic Strings	` ` (backticks – for template literals <code>\${var}</code> – overhead)
JS Multi-line Strings	` ` (backticks – to remove need for <code>\n</code>)

Note: [8:58:46 AM 20-Mar-2025] The **ONLY** time “ ” should be used is for strings **WITHIN** JavaScript strings. Single ‘ within JS Strings should always be escaped: \', backslash should always be escaped: \\



Functions

- Always use Named Function expressions. ([Code Consistently](#), [Code Explicitly](#)).
- **Note:** It helps when viewing stack-traces during debugging.

```
// anonymous function expression
var anonymous = function()
{
    return true;
};

// named function expression
var named = function named()
{
    return true;
};
```

- Never name a parameter “arguments”. This will take precedence over the arguments object that is given to every function scope. ([Code Explicitly](#)).

```
// bad
function dontDoThis(name, options, arguments) {
    // ...stuff...
}

// good
function thisIsOk(name, options, qualities)
{
    // ...stuff...
}
```

Arrow Functions

- Always code ‘fat arrow’ functions the same way, with Aligned Braces `{ }` and a `return` statement. ([Code Consistently](#), [Code Explicitly](#)).

```
const x = (x, y) => x * y; // bad - removal of option return and braces
const x = (x, y) => { return x * y }; // bad - single line exception
const x = (x, y) =>
{
    return x * y // good - Code Consistently, Code for Readability
};
```



Properties

- Use **dot** (.) notation when accessing properties by name. ([Code Simply](#), [Code Consistently](#), [Code for Readability](#)).

```
var luke = {  
  jedi: true,  
  age: 28  
};  
  
// bad  
var isJedi = luke['jedi'];  
  
// good  
var isJedi = luke.jedi;
```

- Use **bracket notation** [] when accessing properties with a variable. ([Code Explicitly](#)).

```
var luke = {  
  jedi: true,  
  age: 28  
};  
  
function getProp(prop)  
{  
  return luke[prop];  
}  
  
var isJedi = getProp('jedi');
```



Variables

- Always use `const` or `let` to declare variables. Not doing so will result in global variables. We want to avoid polluting the global namespace. Globals should only be used for app 'context', or 'common memory', usually loaded from a preferences files, settings files, or app 'memory' files. ([Code Explicitly](#)).

```
// bad - automatic global created
superPower = new SuperPower();

// good - local and 'readonly'
const superPower = new SuperPower();
```

- Use one `const` or `let` declaration per variable or assignment. ([Code Simply](#), [Code Explicitly](#)).

Why? It's easier to add new variable declarations this way, and you never have to worry about swapping out a `;` for a `,` or introducing punctuation-only diffs. You can also step through each declaration with the debugger, instead of jumping through all of them at once.

```
// bad
const items = getItems(),
      goSportsTeam = true,
      dragonball = 'z';

// bad
// (compare to above, and try to spot the mistake)
const items = getItems(),
      goSportsTeam = true;
      dragonball = 'z';

// good
const items = getItems();
const goSportsTeam = true;
const dragonball = 'z';
```



- Group all your **const** and then group all your **let**. ([Code for Readability, Code Consistently](#)).
- See **Appendix MicroCODE JS Code Ordering under JS Class Structure**

Why? This is helpful later on you might need to assign a variable depending on one of the previously assigned variables.

```
// bad
let i, len, dragonball,
    items = getItems(),
    goSportsTeam = true;

// bad
let i;
const items = getItems();
let dragonball;
const goSportsTeam = true;
let len;

// CONSTANTS -- good
const goSportsTeam = true;
const items = getItems();

// FIELDS -- good
let dragonball;
let i;
let length;
```

MCODE Style eliminates



Comparison Operators & Equality

- Use `===` and `!==` over `==` and `!=`. ([Code Explicitly](#)).
- Conditional statements such as the `if` statement evaluate their expression using coercion with the `.ToBoolean` abstract method and always follow these simple rules:
 - **Objects** evaluate to **true**
 - **Undefined** evaluates to **false**
 - **Null** evaluates to **false**
 - **Booleans** evaluate to **the value of the boolean**
 - **Numbers** evaluate to **false** if **+0, -0, or NaN**, otherwise **true**
 - **Strings** evaluate to **false** if an empty string "", otherwise **true**

```
if ([0]) {  
  // true  
  // An array (even an empty one) is an object, objects evaluate to true  
}
```

- Use shortcuts for **Booleans**, but explicit comparisons for strings and numbers. For more information see [Truth Equality and JavaScript](#) by Angus Croll. ([Code Simply](#), [Code Explicitly](#)).

```
// bad - boolean  
if (isValid === true) {  
  // ...stuff...  
}  
  
//good - boolean  
if (isValid)  
{  
  //...stuff  
}  
  
// bad - string  
if (name) {  
  //...stuff  
}  
  
// good - string  
if (name !== '') or if (name.length > 0)  
{  
  //...stuff  
}  
  
// bad - array  
if (collection.length) {  
  //...stuff  
}  
  
// good - array  
if (collection.length > 0)  
{  
  //...stuff  
}
```



Blocks

- ~~Use~~ Always use code blocks in braces ~~with all multi-line blocks~~ for conditionals. (Code Consistently).
- See **MicroCODE: Indents, Braces, Blocks, and Aligned-Code**.

```
// bad
if (test)
  return false;

// bad
if (test) return false;

// good
if (test)
{
  return false;
}

// bad
function () { return false; }

// good
function ()
{
  return false;
}
```

MCODE Style eliminates

- ~~If you're using~~ When using ~~multi-line blocks with if~~ and ~~else~~, put ~~else on the same line as your if block's closing brace~~ on a new line, aligned with the ~~if~~. (Code Consistently).
- See **MicroCODE: Indents, Braces, Blocks, and Aligned-Code**.

```
// bad
if (test) {
  thing1();
  thing2();
} else {
  thing3();
}

// good - this will always be easier to read and maintain than the above
if (test)
{
  thing1();
  thing2();
}
else
{
  thing3();
}
```

MCODE Style eliminates



Commas

- Leading commas: **Nope.** ([Code for Readability](#)).
- What barbarian would ever do this?

```
// bad
var story = [
  once
  , upon
  , aTime
];

// good
var story =
[
  once,
  upon,
  aTime
];

// bad
var hero = {
  firstName: 'Bob'
  , lastName: 'Parr'
  , heroName: 'Mr. Incredible'
  , superPower: 'strength'
};

// good
var hero =
{
  firstName: 'Bob',
  lastName: 'Parr',
  heroName: 'Mr. Incredible',
  superPower: 'strength'
};
```



- Additional trailing comma: **Nope**. This can cause problems with IE6/7 and IE9 if it's in quirksmode. Also, in some implementations of ES3 would add length to an array if it had an additional trailing comma. This was clarified in ES5 ([source](#)):

Edition 5 clarifies the fact that a trailing comma at the end of an Array Initializer does not add to the length of the array. This is not a semantic change from Edition 3, but some implementations may have previously misinterpreted this.

- While it's tempting to leave the last comma to making adding future items easier the last trailing comma misleads the compiler/interpreter, and you never want to do that. ([Code Explicitly](#)).

```
// bad
var hero = {
  firstName: 'Kevin',
  lastName: 'Flynn',
};

var heroes = [
  'Batman',
  'Superman',
];

// good
var hero =
{
  firstName: 'Kevin',
  lastName: 'Flynn'
};

var heroes =
[
  'Batman',
  'Superman'
];
```



Semicolons

- **Yup.** ([Code Explicitly](#)).
- **Do not rely on Automatic Semicolon Insertion (ASI).**
- Just get used to semicolons, otherwise you are relying on the compiler/interpreter to figure out what you wanted across multiple lines of code. And sometimes it will make a different decision than you did.

```
// bad - raises exception
const luke = {}
const leia = {}
[luke, leia].forEach((jedi) => jedi.father = 'vader')
```

// bad - raises exception

```
const reaction = "No! That's impossible!"
(async function meanwhileOnTheFalcon() {
  // handle `leia`, `lando`, `chewie`, `r2`, `c3p0`
  // ...
})();
```

// bad - returns `undefined` instead of the value on the next line - always happens when
`return` is on a line by itself because of ASI!

```
function foo() {
  return
  'search your feelings, you know it to be foo'
}
```

// good

```
const luke = {};
const leia = {};
[luke, leia].forEach((jedi) => {
  jedi.father = 'vader';
});
```

// good

```
const reaction = "No! That's impossible!";
(async function meanwhileOnTheFalcon() {
  // handle `leia`, `lando`, `chewie`, `r2`, `c3p0`
  // ...
})();
```

// good

```
function foo() {
  return 'search your feelings, you know it to be foo';
}
```



Naming Conventions

- Avoid single letter names. Be descriptive with your naming. ([Code Explicitly](#), [Code for Readability](#)).

```
// bad
function q() {
  // ...stuff...
}

// good
function sizeQuery()
{
  // ..stuff..
}
```

- Use **camelCase** when naming **objects**, **functions**, and **instances**. ([Code for Readability](#)).

```
// bad
var OBJEcttsssss = {};
var this_is_my_object = {};
var o = {};
function c() {}

// good
var thisIsMyObject = {};
function thisIsMyFunction() {}
```

- Use **PascalCase** when naming **classes** or **components**. ([Code for Readability](#)).

```
// bad
class user {
  constructor(options) {
    this.name = options.name;
  }
}

var bad = new user({
  name: 'nope'
});

// good
class User
{
  constructor(options)
  {
    this.name = options.name;
  }
}

var user = new User(
{
  name: 'yup'
});
```



- Name your functions. This is helpful for stack traces. ([Code Explicitly](#), [Code for Readability](#)).
- **Note:** IE8 and below exhibit some quirks with named function expressions. See <http://kangax.github.io/nfe/> for more info.

```
// bad
var log = function (msg) {
  console.log(msg);
};

// good
var log = function log(msg)
{
  console.log(msg);
};
```

- Start PRIVATE METHOD names within a Class with an '_' (Underscore).
- And document the method as `@api private` in the method header.

```
// METHODS - PRIVATE
/**
 * _method1() - description of private method.
 *
 * @param1 {type} description of param1.
 * @returns {type} description of return value.
 * @api private
 */
_method1(param1)
{
  // ...

  return value;
}
```

- Start all PRIVATE FIELD names within a Class with a '#' (Pound Sign / Hash).

```
// PRIVATE FIELDS
#property1 = 0.00;
#property2 = '';

// CONSTRUCTOR
constructor(objectName)
{
  this.type = CLASS_TYPE;
```

- Make all CONSTANTS UPPERCASE and **static**

```
// CONSTANTS
static MIN_VALUE = 1;
static MAX_VALUE = 999;
static CLASS_TYPE = 'Example';
```



Common Patterns

Note: Value comparisons, test variable against a limit, not a limit against a variable...
(This is known as a '**Yoda Conditional**' and is like saying "if blue is the sky"). ([Code for Readability](#)).

Use...

```
if (ObjectBeingTested == TestValue)
{
    ...
}
```

Not...

```
if (TestValue == ObjectBeingTested)
{
    ...
}
```

Example:

Use...

```
if (Object == null)
{
    ...
}
```

Not...

```
if (null == Object)
{
    ...
}
```

Reason:

The habit of placing the Object being tested first matches the way we speak, "If the object is null" and it makes ordered comparisons logically correct like this...

Use...

```
if (Object >= LowLimit)
{
    ...
}
```

Not...

```
if (LowLimit <= Object)
{
    ...
}
```



Objects and Database Naming

Note: Following one predictable pattern when naming Objects stored in Databases, and the Modules and Methods/Functions used for their Create, Read, Update, and Delete (CRUD) makes code maintenance and readability far easier. ([Code for Readability](#)).

The naming we standardize is comprised of:

- Object Names
- Database Tables
- Object Ids
- Database Object Variables
- Database Object Access

Object Names

Objects stored in a database should have the simplest unique name possible, e.g.: **user** for the users of a system, **account** for accounts containing users.

Object names should group related objects naturally, e.g.: **plc**, **plc_family**, **plc_mfg**, etc.

Where **plc** is the primary object, **plc_mfg** is a list of manufacturers referenced by the plc object.

Database Tables

Database Tables should be named for the object they store in the singular, e.g.: user objects are stored in the **user** table, not a 'users' table.

Object Ids

The unique Identifier of all objects shall be named **_id** within its table or document.

References to that unique id from another object will always be: <objectName>**_id**.

The unique Id of any Object stored in a database should be the native id generated by the database itself, preferable a Universally Unique Identifier (**UUID**) – a standard UUID is made up of hex digits and includes 32 hexadecimal characters displayed in five groups separated by hyphens, in the form of 8-4-4-4-12. This is a total of 36 characters (32 alphanumeric characters and 4 hyphens),
e.g.: UUID: 12345678-1234-1234-1234-123456789ABC = 36 char.

If the database's own unique ID is not a proper UUID, replace the database's default do not add a 2nd Id of your own, e.g.: MongoDB generates a default **_id**: 65b2efc5b9835d9fc151ca01 = 24 char (no dashes).



Example: When using MongoDB generate a 32 char UUID as the database's `_id`.

user	account	email	event	feedback	key	log	usage
_id	name	email					
696a3d36492b41229cd59770bb155868	Abby I. Normal	admin@mcode.com					\$2b\$10\$3d
351a60ce4eb94884816e48eb7b9b5b4c	TIMMMY	tim.mcguire@mcode.com					\$2b\$10\$3K

```

/* SCHEMA
 * user.*
 */
const userSchema = new Schema([
  /* id: {type: String, required: true, unique: true}, // [2024-01-24: TJM] use UUID as MongoDB's "_id"
  _id: {type: String, default: () => uuidv4().replace(/-/g, '')},
  name: {type: String, required: true},
  email: {type: String, required: true},
  password: {type: String},
  avatar: {type: String},
  settings: {type: Object, required: true},
  tfa_enabled: {type: Boolean, required: true},

```

Object Variables

When an object is retrieved from the database the variable used in the code shall match the name of the Object itself, e.g.: when retrieving a **user** from the **user** table the variable loaded shall be named **user**.

```

/* C[R]UD
 * user.get()
 * get a single user
 */
0 references | 0 references
exports.get = async function (req, res)
{
  const user_id = req.params._id || req.user_id;
  utility.assert(user_id, 'Please log in or specify a User Id');

  const user = await userModel.get({_id: user_id, account_id: req.account_id});
  user.accounts = await userModel.accounts({_id: user_id});

  if (req.permission === 'owner')

```

Database Columns					
App type	Example	Data type	Size		Format as stored
<action>_at	active_at	TIMESTAMP	32		A timestamp of the <action>
					UUID: 12345678-1234-1234-1234-123456789ABC = 36 char
NOTE: Using a UUID as "_id" directly from MongoDB as "id". Generated by a default function in all Schema.					
_id	_id	UUID	32		A UUID identifying an Object in its Table. The "id" column name will "not" be used for any other purpose.
<object>_id	user_id	UUID	32		A UUID reference to <object>. The "_id" suffix will "not" be used for any other purpose.
key	[2]	INT	64K		A simple number uniquely identifying a Table row. Only used for internal App data, like email templates, "not" for pri User data.
<object>_ident	user_ident	TEXT	255		A Object's Unique Identity in an external System, "not" a local DB "id".
					Search for "Schema = new Schema(" -- to check create of records against definition changes



Model, View, Controller (MVC) Naming

Note: Following one predictable pattern when naming Objects stored in Databases, and the Modules and the Methods/Functions used for their Create, Read, Update, and Delete (CRUD) makes code maintenance and readability far easier. ([Code for Readability](#)).

First a mnemonic: There is a **MODEL** and a **VIEW**, and they are connected by a **CONTROLLER**.

Input: View → Controller → Model
Output: Model → Controller → View

The naming we standardize is comprised of:

- **Model:** Represents the data and business logic of the application. It manages the data, logic, and rules of the application and notifies the view and controller of any changes.
- **View:** Represents the presentation layer of the application. It displays data to the user that is provided by the model and sends user commands to the controller.
- **Controller:** Acts as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the Model component and interact with the Views to render the final output.

Model

The module holding an Object's Model code shall be named: <object>.model.js
e.g.: **user.model.js**

Variables accessing the Model shall be named: <object>**Model**
e.g.: **userModel**

View

The module holding an Object's View code shall be named after the Object itself: <object>.js
e.g.: **user.jsx** (React UI Component)

Variables accessing the View are React Components shall have Pascal Names: <Object>
e.g.: **User**

A module holding a View of multiple Objects shall be named: <object>s.js
e.g.: **users.jsx** (React UI Component)

Variables accessing the View are React Components shall have Pascal Names: <Object>s
e.g.: **Users**

Controller

The module holding an Object's Controller code shall be named: <object>.controller.js
e.g.: **user.controller.js**

Variables accessing the Model shall be named: <object>**Controller**
e.g.: **userController**



Code Files					
Folder	Example	Type	File Extension		Description
CLIENT					
component \	form.jsx	object	*.jsx		VIEW - The React App Component that visualizes 'object'.
component \	form.tailwind.js	object	*.tailwind.js		VIEW - The styling used to visualize 'object'. Option #1
component \	form.module.scss	object	*.module.scss		VIEW - The styling used to visualize 'object'. Option #2
SERVER					
api \	user.route.js	object	*.route.js		ROUTE - API (Express) routing from VIEW to CONTROLLER for 'object'.
controller \	user.controller.js	object	*.controller.js		CONTROLLER - the business logic for 'object' and the CRUD connection to the MODEL.
model \	user.model.js	object	*.model.js		MODEL - the Database connection / storage / retrieval for 'object' to handle MODEL requests.
test \	user.test.js	object	*.test.js		TEST - test code for 'object' to be exercised in CI-CD pipeline.



Create, Read, Update, Delete (CRUD) Naming

Note: Following one predictable pattern when naming Methods and Functions used to access a Database makes code maintenance and readability far easier. ([Code for Readability](#)).

The naming we standardize is comprised of:

- **Create:** The Object Model Function which adds a new Object in a Database Table.
- **Read:** The Object Model Function which retrieves an Object from a Database Table.
- **Update:** The Object Model Function which modifies an Object in a Database Table.
- **Delete:** The Object Model Function which removes an Object from a Database Table.

Create

The function is held in the module holding an Object's Model code named: `<object>.create()`
e.g.: `user.create` = async function (req, res)

This **Model** function will be accessed from the API **Route** by HTTP **POST** thru the **Controller** function of the exact same name.

```
api.post('/api/<object>', middleware(<args>), use(<object>Controller.create));
```

Read

The function is held in the module holding an Object's Model code named: `<object>.get()`
e.g.: `user.get` = async function (req, res)

This **Model** function will be accessed from the API **Route** by HTTP **GET** thru the **Controller** function of the exact same name.

```
api.get('/api/<object>', middleware(<args>), use(<object>Controller.get));
```

Update

The function is held in the module holding an Object's Model code named: `<object>.update()`
e.g.: `user.update` = async function (req, res)

This **Model** function will be accessed from the API **Route** by HTTP **PATCH** or **PUT** thru the **Controller** function of the exact same name.

```
api.patch('/api/<object>', middleware(<args>), use(<object>Controller.update));
```



Delete

The function is held in the module holding an Object's Model code named: `<object>.delete()`
 e.g.: `user.delete` = async function (req, res)

This **Model** function will be accessed from the API **Route** by HTTP **DELETE** thru the **Controller** function of the exact same name.

```
api.delete('/api/<object>/:id', middleware(<args>), use(<object>Controller.delete));
```

MongoDB Access Vars					
Name	Example	Data type	Source	Usage	Description
req	<code>req</code>	Express.Request	API	Controller	Request from the API.
res	<code>res</code>	Express.Response	Controller	API	Responses to the API.
data	<code>data</code>	JSON	any	any	Temporary data while an Object is being constructed, e.g.: <code>data</code> contains a fields of a record .
<doc>	<code>user</code>	JSON	Model	Controller, Mongoose	Mogo Document . An Object as stored in the MongoDB (i.e.: a Mongo 'Document'), e.g.: <code>userData</code> is a complete record from the <code>user</code> table.
<doc>Model	<code>userModel.get()</code>	JavaScript	Model	Controller	App Model (CRUD Layer) as referenced in the Controller , e.g.: <code>user.Model.js</code> is the User Model code.
<doc>Controller	<code>userController.get()</code>	JavaScript	Model	Route	App Controller (API Layer) as referenced in the Route , e.g.: <code>user.Controller.js</code> is the User Controller code.
<doc>Schema	<code>userSchema</code> <code>user.schema</code>	Schema()	Mongoose	MongoDB	Mongo Document Definition . The definition of table record data (i.e.: a Mongo 'document'), e.g.: <code>userSchema</code> defines all the data in a record from the <code>user</code> table.
<Doc>	<code>User.find()</code> <code>user.model.find()</code>	JavaScript	MongoDB	Model	Mongoose Model (Database Class/Constructor Layer). The direct access to a table (i.e.: a Mongo 'collection'), in the database, e.g.: <code>User</code> accesses the <code>user</code> table in MongoDB.



Code Maintenance

Follow a consistent procedure for editing released code. ([Code Consistently](#)).

1 – Follow a 'Quality Control Checklist' as you build and document a Release or Hot Fix.

Don't wait until you are done to 'pencil whip' the form, it is meant to be used as you go through the process.

 MicroCODE App Release Process Quality Control Checklist	
Background This defines the steps to successfully release MicroCODE Apps to Clients. App: Control (EPP) Build: v2.0.1 b (9)	■ STEP 6a: Document--with user requirements--the NEW FEATURES being added. These go into the RELEASE NOTES at the end under "New in Version vM.m.R c (B)" Document the Test Cases. ■ STEP 6b: Document--with screen shots if helpful--the CORRECTED ISSUES being resolved. These go into the RELEASE NOTES at the end under "Correct in Version vM.m.R c (B)" Document the Test Cases. NOTE: Both 'New' and 'Corrected' go into the Release Notes regardless. If nothing changed note "No features were added" or "No issues were corrected".
PREPARATION ■ STEP 1: Make a copy of the most recent INTERNAL source code tree. <u>Do not edit a stored copy of the 'Internal' source code tree.</u> ■ STEP 2: Rename it 'D:\MicroCODE\Internal' -- all coding tools reference this path.	

The Checklist should include documentation templates for the README.md, Release Notes and Test Cases...

MicroCODE App Release Process Quality Control Checklist 	
New in Version vM.m.R c (B) The following features were added in this Release: 1) Requirement: ... Implementation: ... TEST CASES: These software tests were performed prior to release to ensure App functionality. Test Case N: ... – Passed. Test Case N: ... – Conditional. Test Case N: ... – Failed.	Corrected in Version VM.m.R c (B) The following defects were fixed this Release: 1) Issue: ... Correction: ... TEST CASES: These software tests were performed prior to release to ensure App functionality. Test Case N: ... – Passed. Test Case N: ... – Conditional. Test Case N: ... – Failed.



Appendix A: MicroCODE JS Class Structure

According to the [ES6 Rules Documentation](#)—and most JavaScript Style Guides—the ordering of the Class elements is shown below. This is the ordering enforced by the MicroCODE JS Class Template. ([Code Explicitly](#), [Code for Readability](#), [Code for Readability](#)).

Within a **class** (`ClassName`) group **datatypes** in this order:

- Constant Variables/Fields (`const`)
- Fields (`let`)
- Constructor (`constructor`)
- Enums (`const Object.freeze({})`)
- Properties (`get`, `set`)
- Iterators (`[Symbol.*]`)
- Methods (`methodName(param)`)
- Nested Classes

Within each of **type** group order by **access**:

- public
- private

Within each of the **access** groups, order by **static**, then **non-static**:

- static
- non-static

Within each of the **static/non-static** groups of fields, order by **readonly**, and then **non-readonly**:

- readonly
- non-readonly

e.g.: The methods section then would be unrolled in this order:

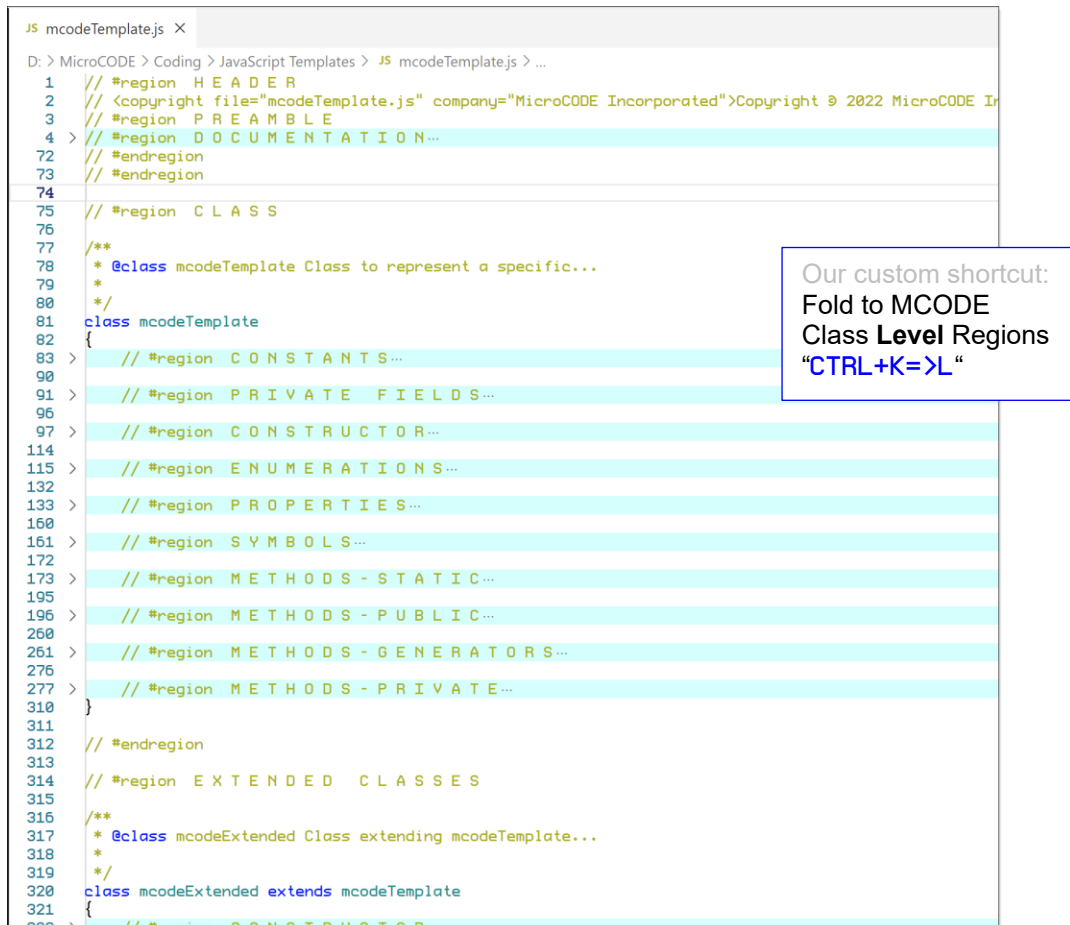
- public static methods
- public methods
- private static methods
- private methods



JS Class Structure (MicroCODE)

`mcodeTemplate` is technically a function (the one that we provide as constructor), while methods, getters and setters are written to `mcodeTemplate.prototype`. Below is our actual JavaScript Class Template file used to start all new JS Classes in our projects. ([Code Consistently](#)).

This template is built with **Code Folding** support through use of “`// #region`” and “`// #endregion`”. This will seem like a lot of unnecessary ‘syntactic sugar’—I hate that phrase BTW—reading through the following and understanding what it enables in VS CODE should make the value clear. Start with the entire file ‘folded’ to Class level. “`CTRL+K=>3`”.



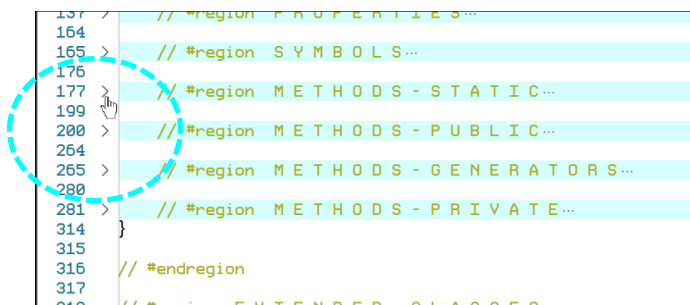
```

JS mcodeTemplate.js X
D: > MicroCODE > Coding > JavaScript Templates > JS mcodeTemplate.js > ...
1 // #region H E A D E R
2 // <copyright file="mcodeTemplate.js" company="MicroCODE Incorporated">Copyright © 2022 MicroCODE Inc
3 // #region P R E A M B L E
4 // #region D O C U M E N T A T I O N ...
72 // #endregion
73 // #endregion
74
75 // #region C L A S S
76
77 /**
78  * @class mcodeTemplate Class to represent a specific...
79  *
80  */
81 class mcodeTemplate
82 {
83 > // #region C O N S T A N T S ...
90
91 > // #region P R I V A T E   F I E L D S ...
96
97 > // #region C O N S T R U C T O R ...
114
115 > // #region E N U M E R A T I O N S ...
132
133 > // #region P R O P E R T I E S ...
160
161 > // #region S Y M B O L S ...
172
173 > // #region M E T H O D S - S T A T I C ...
195
196 > // #region M E T H O D S - P U B L I C ...
260
261 > // #region M E T H O D S - G E N E R A T O R S ...
276
277 > // #region M E T H O D S - P R I V A T E ...
310
311 }
312 // #endregion
313 // #region E X T E N D E D   C L A S S E S
314
315 /**
316  * @class mcodeExtended Class extending mcodeTemplate...
317  *
318  */
319
320 class mcodeExtended extends mcodeTemplate
321 {
322 > // #region C O N S T R U C T O R

```

Our custom shortcut:
Fold to MCODE
Class **Level** Regions
“`CTRL+K=>L`”

Then ‘unfold’ the area you are working my clicking the “`>`” for that Region.



```

157 > // #region P R O P E R T I E S ...
164 > // #region S Y M B O L S ...
176 > // #region M E T H O D S - S T A T I C ...
177 > // #region M E T H O D S - P U B L I C ...
199 > // #region M E T H O D S - G E N E R A T O R S ...
200 > // #region M E T H O D S - P R I V A T E ...
264 > // #region M E T H O D S - G E N E R A T O R S ...
265 > // #region M E T H O D S - P R I V A T E ...
280 > // #region M E T H O D S - P R I V A T E ...
281 > // #region M E T H O D S - P R I V A T E ...
314 }
315 // #endregion
316 // #region E X T E N D E D   C L A S S E S
317

```

This leaves everything you are not working on 'hidden' via the 'folded code' support in VS Code.

```

118 >
119 > // #region ENUMERATIONS...
136
137 > // #region PROPERTIES...
164
165 > // #region SYMBOLS...
176
177 > // #region METHODS - STATIC
178
179 > /**
180  * static1() - description of public static method, called by prototype not object.
181  *           This does not operate on a specific copy of a Class object.
182  * @api public
183  *
184  * @param {type} param1 description of param1.
185  * @returns {type} description of return value.
186  *
187  * @example
188  *
189  *     static1('param1');
190  */
191 > static static1(param1)
192 {
193     // ...
194
195     return value;
196 }
197
198 // #endregion
199
200 > // #region METHODS - PUBLIC...
264
265 > // #region METHODS - GENERATORS...

```

All the 'code folding' can be opened with... "CTRL+K=>J".

```

JS mcodeTemplate.js x
D:\> MicroCODE > Coding > JavaScript Templates > JS mcodeTemplate.js > mcodeTemplate
57
58 *
59 *
60 *
61 * Date:      By-Group:  Rev:      Description:
62 *
63 * 02-Feb-2022 TJM-MCODE {0001} New module for common reusable Javascript Classes for code
64 * 06-Feb-2022 TJM-MCODE {0002} Moved @api tag under method description and before @param
65 * 08-Feb-2022 TJM-MCODE {0003} Added @constructor, @property, @class, @enum,
66 *
67 *
68 */
69 "use strict";
70
71 // #endregion
72 // #endregion
73 // #endregion
74
75 // #region CLASS
76
77 /**
78  * @class mcodeTemplate Class to represent a specific...
79  *
80  */
81 class mcodeTemplate
82 {
83     // #region CONSTANTS
84
85     static MIN_VALUE = 1;
86     static MAX_VALUE = 999;
87     static CLASS_TYPE = 'Example';
88
89     // #endregion
90
91     // #region PRIVATE FIELDS
92     #property1 = 0.00;
93     #property2 = '';
94
95     // #endregion
96
97     // #region CONSTRUCTOR

```

Standard shortcut:
Unfold ALL Regions
"CTRL+K=>J"



Everything can be 'folded' to LEVEL 0 with... "CTRL+K=>0".

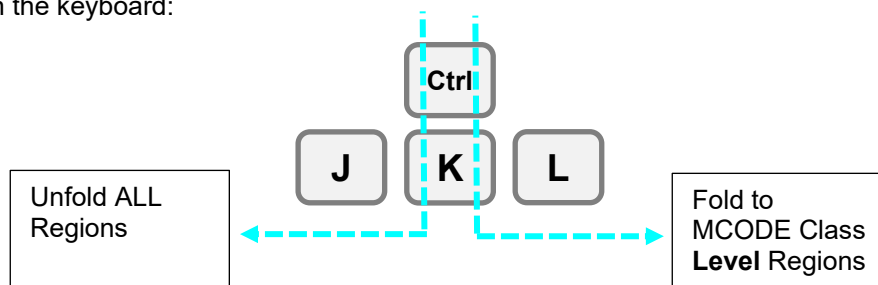
And you can restore your view of the overall Class structure (locked by **MCODE Style**) at any time with...

"CTRL+K=>J" "CTRL+K=>3".

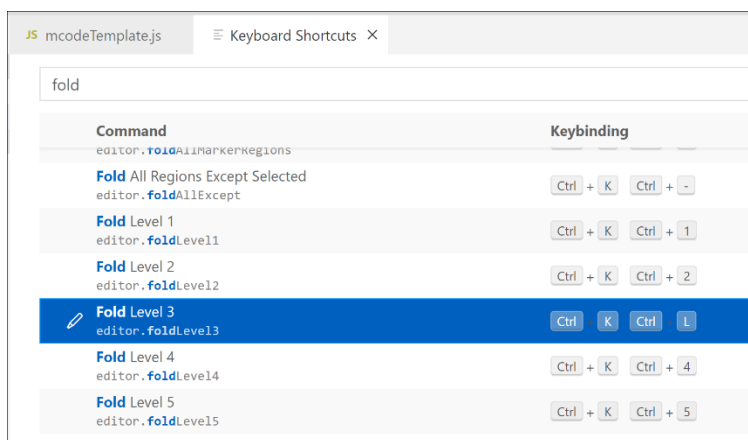
NOTE: You cannot get the same view with "CTRL+K=>3" without starting with "CTRL+K=>J". These commands are 'relative' to the current view, not 'absolute'... i.e.: "CTRL+K=>3" will not always give you the same result by itself.

I use this all day every day when navigating files, so I have remapped "CTRL+K=>L" to equal "CTRL+K=>3".

So, on the keyboard:



- File → Preferences → Keyboard Shortcuts



```
// #region H E A D E R
// <copyright file="mcodeTemplate.js" company="MicroCODE Incorporated">Copyright © 2022 MicroCODE Incorporated Troy,
MI</copyright><author>Timothy J. McGuire</author>
// #region P R E A M B L E
// #region D O C U M E N T A T I O N
/*
 * Title:      MicroCODE Common JavaScript Class Template
 * Module:     Modules (MicroCODE:mcodeTemplate.js)
 * Project:    MicroCODE Common Library
 * Customer:   Internal
 * Creator:    MicroCODE Incorporated
 * Date:       February 2022
 * Author:     Timothy J McGuire
 *
 * Designed and Coded: 2022 MicroCODE Incorporated
 *
 * This software and related materials are the property of
 * MicroCODE Incorporated and contain confidential and proprietary
 * information. This software and related materials shall not be
 * duplicated, disclosed to others, or used in any way without the
 * written of MicroCODE Incorporated.
 *
 * DESCRIPTION:
 * -----
 *
 * This module implements the MicroCODE's Common JavaScript Class Template.
 * This file is copied to start all MicroCODE JavaScript code files.
 *
 * REFERENCES:
 * -----
 *
 * 1. MIT xPRO Style Guide
 *    https://student.emeritus.org/courses/3291/files/2554233/download?wrap=1
 *
 * 2. AirBnB JavaScript Style Guide
 *    https://github.com/airbnb/javascript
 *
 * 3. Turing School Style Guide
 *    https://github.com/turingschool-examples/javascript/tree/main/es5
 *
 * 4. MDN Web Docs - JavaScript Classes
 *    https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Classes
 *
 * 5. JSDoc - How to properly document JavaScript Code.
 *    https://
 *
 * 4. MicroCODE MCX-S02 (Internal JS Style Guide).docx
 *
 * DEMONSTRATION VIDEOS:
 * -----
 *
 * 1. ...
 *
 * MODIFICATIONS:
 * -----
 *
 * Date:      By-Group:   Rev:    Description:
 *
 * 02-Feb-2022  TJM-MCODE  {0001}  New module for common reusable Javascript Classes for code files.
 * 06-Feb-2022  TJM-MCODE  {0002}  Moved @api tag under method description and before @param and @return.
 * 08-Feb-2022  TJM-MCODE  {0003}  Aded @constructor, @property, @class, @enum,
 *
 */
"use strict";

// #endregion
// #endregion
// #endregion

// #region C L A S S

/**
 * @class mcodeTemplate Class to represent a specific...
 */
class mcodeTemplate
{
    // #region C O N S T A N T S

    static MIN_VALUE = 1;
    static MAX_VALUE = 999;
    static CLASS_TYPE = 'Example';

```

```

// #endregion

// #region PRIVATE FIELDS
@property1 = 0.00;
@property2 = '';

// #endregion

// #region CONSTRUCTOR

/**
 * @constructor mcodeTemplate class constructor.
 *
 * @param {string} objectName the name of the object.
 */
constructor (objectName)
{
    this.type = CLASS_TYPE;
    this.name = objectName;
    this.enums = [];

    // ...
}

// #endregion

// #region ENUMERATIONS

/**
 * @enum namedEnum1 - a description of this enum, its use, and meaning.
 */
static namedEnum1 = Object.freeze
(
{
    name1: 0,
    name2: 1,
    name3: 2,
    name4: 3,
    name5: 4,
    name6: 5,
    name7: 6
}
));

// #endregion

// #region PROPERTIES

/**
 * @property {type} property1 a description of this property, its use, and meaning.
 */
get property1()
{
    return this.#property1;
}
set property1(value)
{
    this.#property1 = value;
}

/**
 * @property {type} property2 a description of this property, its use, and meaning.
 */
get property2()
{
    return this.#property2;
}
set property2(value)
{
    this.#property2 = value;
}

// #endregion

// #region SYMBOLS

/**
 * iterator1 - a description of this iterator, its use, and meaning.
 */
[Symbol.iterator]()
{
    // method with computed name (symbol here)
}

// #endregion

// #region METHODS - STATIC

/**
 * static1() - description of public static method, called by prototype not object.
 * This does not operate on a specific copy of a Class object.

```



```

* @api public
* @param {type} param1 description of param1.
* @returns {type} description of return value.
* @example
*     static1('param1');
*/
static static1(param1)
{
    // ...

    return value;
}

// #endregion

// #region METHODS - PUBLIC
/**
 * method1() - description of public method.
 * @api public
 * @param {type} param1 1st method parameter.
 * @returns method result.
 * @example
 *     method1('param1');
 */
method1(param1)
{
    // ...

    return value;
}

/**
 * method2() - description of public method.
 * @api public
 * @param {type} param1 1st method parameter.
 * @param {type} param2 2nd method parameter.
 * @returns method result.
 * @example
 *     method2('param1', 'param2');
 */
method2(param1, param2)
{
    // ...

    return value;
}

/**
 * method3() - description of public method.
 * @api public
 * @param {type} param1 1st method parameter.
 * @param {type} param2 2nd method parameter.
 * @param {type} param3 3rd method parameter.
 * @returns {type} method result.
 * @example
 *     method3('param1', 'param2');
 */
method3(param1, param2, param3)
{
    // ...

    return value;
}

// #endregion

// #region METHODS - GENERATORS
/**
 * getValue() - returns all values in 'enums'.
 *
 */

```



```

*getValue()
{
    for (const enumValue of this.enums)
    {
        yield value;
    }
}

// #endregion

// #region METHODS - PRIVATE

/**
 * _method1() - description of private method.    NOTE: Method headers are *optional* for private methods.
 *
 * @api private
 *
 * @param {type} param1 description of param1.
 * @returns {type} description of return value.
 */
method1(param1)
{
    // ...

    return value;
}

/**
 * _method2() - description of private method.
 *
 * @api private
 *
 * @param {type} param1 description of param1.
 * @returns {type} description of return value.
 */
method2(param1)
{
    // ...

    return value;
}

// #endregion
}

// #endregion

// #region EXTENDED CLASSES

/**
 * @class mcodeExtended Class extending mcodeTemplate...
 *
 */
class mcodeExtended extends mcodeTemplate
{
    // #region CONSTRUCTOR

    constructor (name)
    {
        super(name); // call the super class constructor and pass in the name parameter
    }

    // #endregion

    // #region METHODS - PUBLIC

    /**
     * method9() -- ouputs the extended name of the Class.
     *
     */
    method9()
    {
        console.log(`${ this.name } communicates.`);
    }

    // #endregion
}

// #endregion

```



Appendix B: Visual Studio Code Settings

These are the VS Code settings recommended to maintain **MCODE Style** automatically.

- Automatic Spaces

Editor: Insert Spaces

- ☒ Insert spaces when pressing **Tab**. This setting is overridden based on the file contents when [Editor: Detect Indentation](#) is on.

Editor > Comments: Insert Space

- ☒ Controls whether a space character is inserted when commenting.

- When to auto-format

Editor: Format On Paste

- ☒ Controls whether the editor should automatically format the pasted content. A formatter must be available and the formatter should be able to format a range in a document.

Editor: Format On Save

- ☒ Format a file on save. A formatter must be available, the file must not be saved after delay, and the editor must not be shutting down.

Editor: Format On Save Mode

Controls if format on save formats the whole file or only modifications. Only applies when [Editor: Format On Save](#) is enabled.

file



Editor: Format On Type

- ☒ Controls whether the editor should automatically format the line after typing.



- JavaScript auto-formatting options (same for TypeScript)

JavaScript › Format: Enable

☒ Enable/disable default JavaScript formatter.

JavaScript › Format: Insert Space After Comma Delimiter

☒ Defines space handling after a comma delimiter.

JavaScript › Format: Insert Space After Constructor

☒ Defines space handling after the constructor keyword.

JavaScript › Format: Insert Space After Function Keyword For Anonymous Functions

☒ Defines space handling after function keyword for anonymous functions.

JavaScript › Format: Insert Space After Keywords In Control Flow Statements

☒ Defines space handling after keywords in a control flow statement.

JavaScript › Format: Insert Space After Opening And Before Closing Empty Braces

☒ Defines space handling after opening and before closing empty braces.

JavaScript › Format: Insert Space After Opening And Before Closing Jsx Expression Braces

☐ Defines space handling after opening and before closing JSX expression braces.

JavaScript › Format: Insert Space After Opening And Before Closing Nonempty Braces

☒ Defines space handling after opening and before closing non-empty braces.

JavaScript › Format: Insert Space After Opening And Before Closing Nonempty Brackets

☐ Defines space handling after opening and before closing non-empty brackets.

JavaScript › Format: Insert Space After Opening And Before Closing Nonempty Parenthesis

- JavaScript auto-formatting options (continued) (same for TypeScript)

JavaScript › Format: Insert Space After Opening And Before Closing Nonempty Parenthesis

☐ Defines space handling after opening and before closing non-empty parenthesis.

JavaScript › Format: Insert Space After Opening And Before Closing Template String Braces

☒ Defines space handling after opening and before closing template string braces.

JavaScript › Format: Insert Space After Semicolon In For Statements

☒ Defines space handling after a semicolon in a for statement.

JavaScript › Format: Insert Space Before And After Binary Operators

☒ Defines space handling after a binary operator.

JavaScript › Format: Insert Space Before Function Parenthesis

☐ Defines space handling before function argument parentheses.

JavaScript › Format: Place Open Brace On New Line For Control Blocks

☒ Defines whether an open brace is put onto a new line for control blocks or not.

JavaScript › Format: Place Open Brace On New Line For Functions

☒ Defines whether an open brace is put onto a new line for functions or not.

JavaScript › Format: Semicolons

Defines handling of optional semicolons. Requires using TypeScript 3.7 or newer in the workspace.

insert



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MicroCODE CSS Style Guide

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